



Track Removal Tool

D11/D11T

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

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1.0 Introduction

With the increased use of our infrastructure equipment, including the D11 and D11T track tractor, which operate in the S11D mine, there was a need to improve productivity in the activity of changing tracks, mitigating risks of accidents, and reducing the exposure of people during the replacement and interaction with the equipment.

In addition, this tool makes the process of maintaining and replacing the conveyor more practical, as it eliminates the use of chains or cables using the component itself used for the replacement of the new component.

2.0 Best Practice Description

The tool was developed aiming at greater safety of those involved in the activity, since the technicians are no longer exposed to involuntary movements of the tractor, the movement of lifting equipment and projection of materials.

Another important gain is in productivity, with this tool we can change the track with a time gain of approximately 42%, due to greater agility in removal.

3.0 Implementation Steps

For questions regarding the replacement of the track, refer to the Operation and Maintenance Manual, "Track - Check/Adjust" (i03983108).

1º Step: Build and develop the track removal aid tool. Please refer to the attached technical drawing and video or consult the dealer for further information.



Figure 1: Auxiliary conveyor removal tool.

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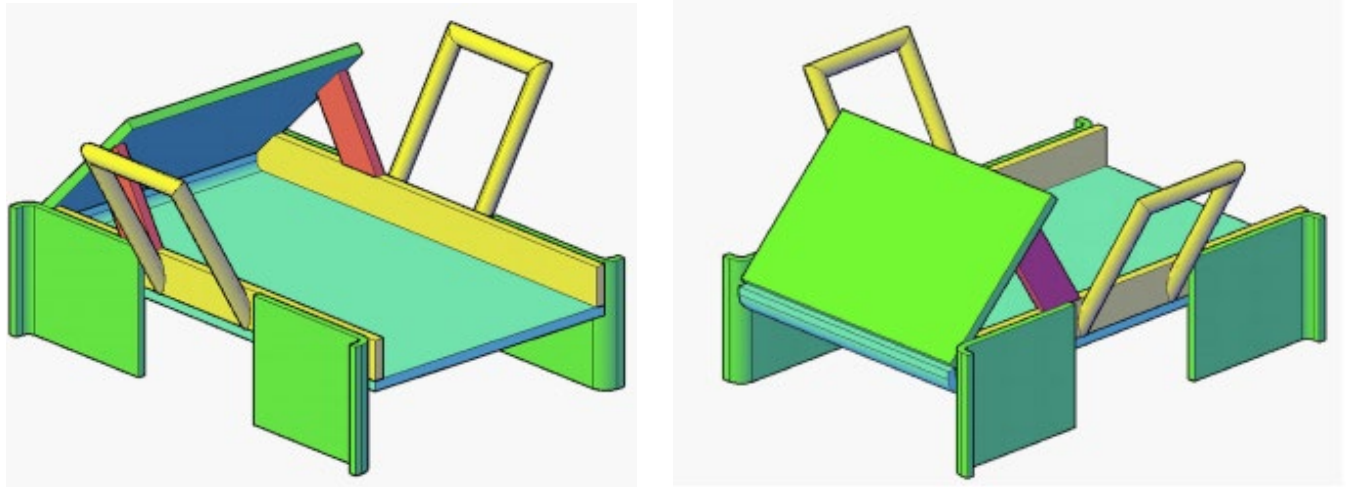


Figure 2: Side view of the auxiliary conveyor removal tool.

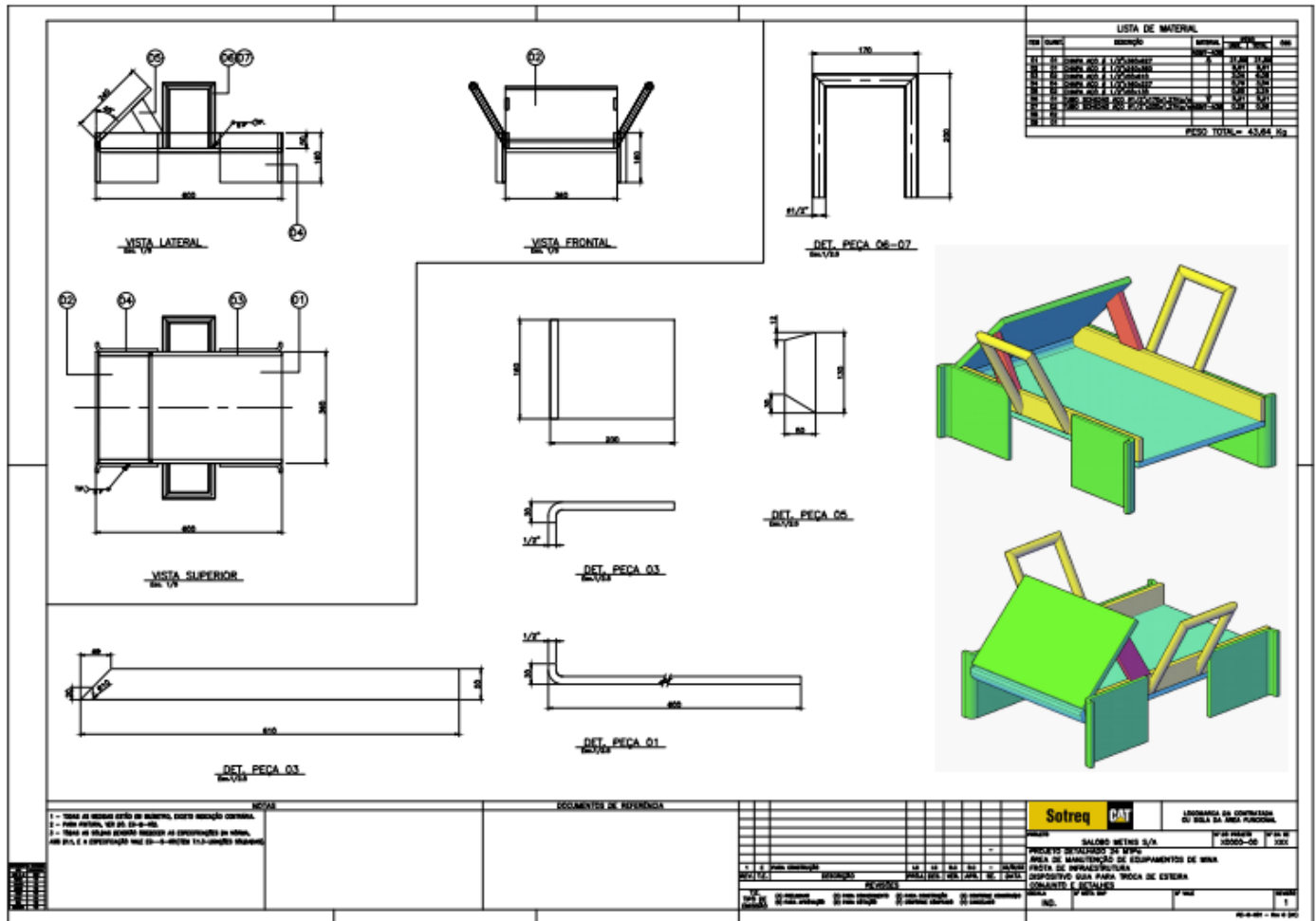


Figure 3: Technical drawing of the auxiliary conveyor removal tool. Measurements in millimeters (mm).

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2° Step: The tool (10) is designed to make the conveyor to be replaced slide onto the new conveyor, facilitating the maneuver of removing and installing the conveyor. This maneuver eliminates the need for forklifts to pull and/or lift the damaged conveyor, as well as eliminates the use of accessories such as cables and/or chains to secure it. As a result, the conveyor remains within the track of the rollers, reducing the risk of material projection.

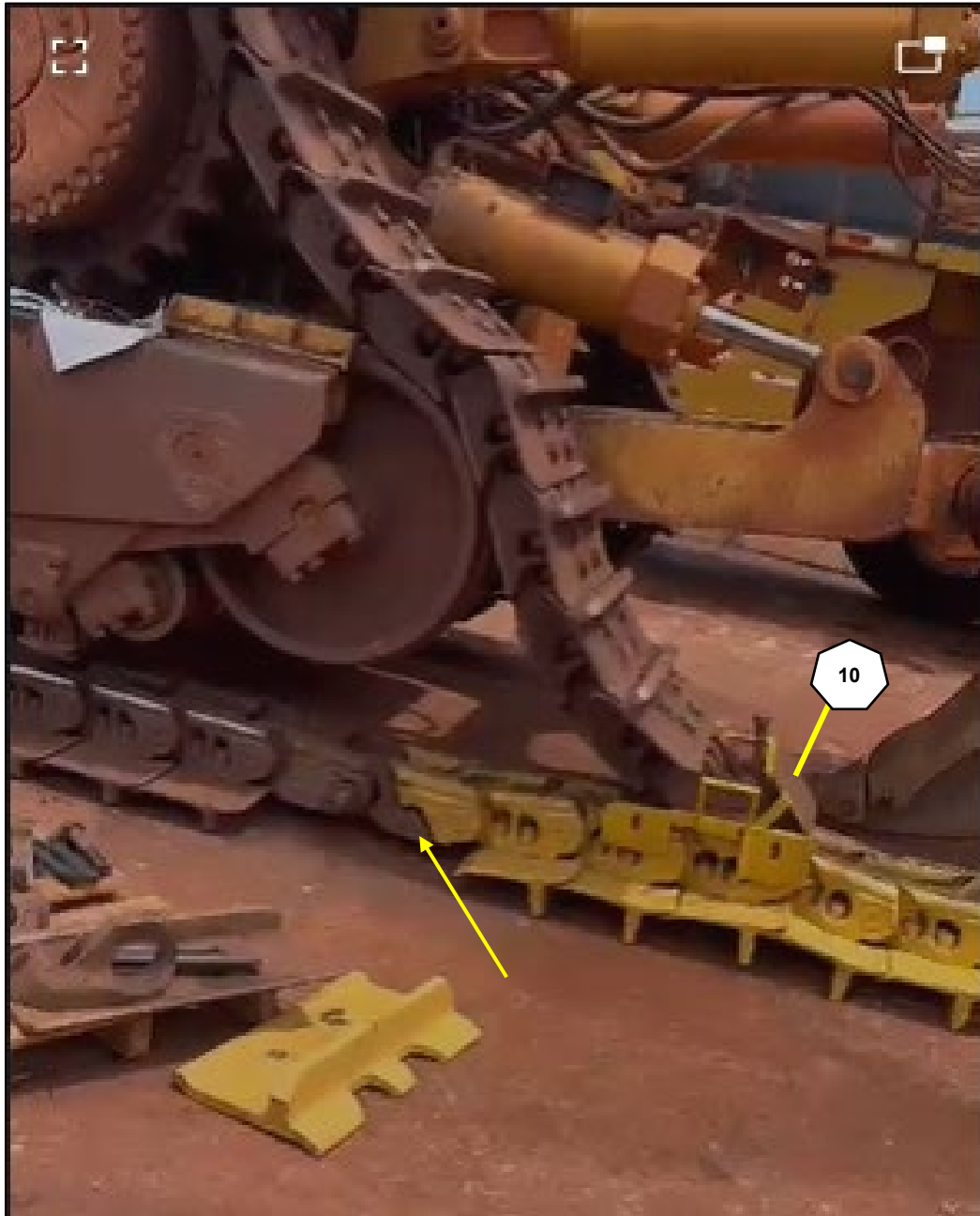


Figure 4: Fitting the ends of the auxiliary conveyor removal tool.

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3° Step: With both ends fitted and the movement of the final control, the damaged conveyor pulls the new conveyor, already positioning it in the ideal position to complete the installation.



Figure 5: Positioning and use of the auxiliary conveyor removal tool.

4° Step: In the movement of removing the damaged track, the track already positioned on the truck and guide wheel, we can position the cradle of the track under the segmented link, positioning to connect the links.



Figure 6: Moving the conveyor along the track.

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4.0 Benefits

- Mitigation between human-machine interaction.
- There is no need to use a forklift and/or crane to reposition the conveyor.
- No exposure of people in the line of fire.
- With the elimination of cables and/or chains to remove the conveyor, the risk of material projection and whipping was eliminated.
- Agility in the replacement of the conveyors.
- With the improvement to remove the conveyor, we gain in time, because the movement of the conveyor with auxiliary feature is no longer necessary. With this the removal becomes a continuous and agile activity.
- Increased productivity.
- Before the tool, we changed a track with approximately 6 hours, meeting all safety requirements. With the new tool, this same activity can be performed in 3.5 hours and with a higher safety factor.

5.0 Resources Required

BILL OF MATERIALS						
ITEM	QUANT.	Description	MATERIAL	Weight		OBS
				UNIT.	TOTAL	
			ASMT-A36			
01	01	STEEL PLATEN # 1/2"x360x627	↗	21,89	21,89	
02	01	STEEL PLATEN # 1/2"x240x360		8,61	8,61	
03	02	STEEL PLATEN # 1/2"x50x610		3,04	6,08	
04	04	STEEL PLATEN # 1/2"x160x227		0,76	3,04	
05	02	STEEL PLATEN # 1/2"x50x130		0,65	3,24	
06	01	STEEL TUBE ø1/2"x170x1,27Kg/m	↘	0,21	0,21	
07	02	STEEL TUBE ø1/2"x200x1,27Kg/m	ASMT-A36	0,28	0,56	

For the manufacturing of the tool, a project was developed in collaboration with the engineering team, and materials found in the client's area were used. We also utilized internal welders for the fabrication of the device, which facilitated and lowered the costs involved in this project. Considering the materials used and the allocated labor, we estimate around \$1,550.00 for manufacturing.

Consult the dealer for further details.

6.0 Supporting Attachments / References

- Forklift 15 tons to position the track in line with the equipment.
- Tool for sliding the track.
- Hand and pneumatic torque tools.
- Drawing and a video can be found in the appendix.

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7.0 Related Best Practices

N/A

8.0 Acknowledgements

SOTREQ (Northern Brazil) would like to express its deep admiration for the professionals involved in improving the maintenance and repair processes developed by Mining Engineering.

SOTREQ would also like to thank the indispensable technical contribution of **Doug Friedrich of Caterpillar Inc.**

This Best Practice was written by:

Daniel Barcelos
Performance Engineer – SOTREQ S/A
daniel.barcelos@sotreq.com.br
+ 55 31 98587-0935

Reinaldo Angelo
Engineering Coordinator - SOTREQ S/A
reinaldo.angelo@sotreq.com.br
+ 55 31 9996-5562

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